

Effat University

Course Syllabus

COURSE TITLE: DEEP LEARNING

COURSE ID: CS4085-2

College:	Engineering	Department:	Computer Science
Semester/Year:	Fall 2026	Prerequisite:	CS 4082
Instructor:	Dr Naila Marir	Office Number:	CoE 108
Course Schedule:	12:00 pm- 1:50 pm UT	Room:	
Office Hours:	Sunday 8:00–10:00 AM- 2:00-4:00 PM; Tuesday 2:00–4:00 pm and Thursday 12:00–4:00 pm		

Course Contact Information

1. Course Instructor Email: namarir@effatuniversity.edu.sa
2. Blackboard Support: BlackboardHelp@effatuniversity.edu.sa
3. General Technical Support: helpdesk@effatuniversity.edu.sa

Welcoming Message

Welcome to the Deep Learning course! This course provides an in-depth introduction to neural networks and modern deep learning. Throughout the semester, we will explore how deep neural networks are built, trained, optimized, and applied to different types of data. Topics include hyperparameter tuning, regularization, optimization, Convolutional Neural Networks (CNNs), sequence models, transformers, generative models, Large Language Models (LLMs), transfer learning, and advanced topics in deep learning. Hands-on laboratories and projects will provide practical experience in developing and evaluating deep learning solutions for real-world AI applications.

Faculty Availability

As the instructor of this course, I will be available to answer any questions or concerns through my email shown above in contacts. Please note that it might take me up to 24 hours to get back to you. Feel free to call me on my office phone during office hours in case you could not meet with me face-to-face.

Students Attendance

Students are required to attend all scheduled class sessions and complete the assigned tasks as outlined in the course plan. Attendance will be recorded from the first class. Attendance grades are distributed equally across all weeks. Missing more than 33% of total class sessions will result in a DN (Denial) grade for the course.

Course Resources and Interactions

- All course materials, including syllabus, lectures, assessments, activities, and rubrics, are available to you on Blackboard and accessible from the first day in your course.
- All announcements related to the course will be available on Blackboard. Please check the course Blackboard often to be aware of all course announcements.
- All course submissions should be on Blackboard or in person, as per the faculty requirement.

Class Participation

Active participation is an essential part of the learning process. Students are strongly encouraged to:

- Ask and answer questions during lectures.
- Contribute to meaningful discussions, both in class and on the course Discussion Board.
- Respond to classmates' ideas in constructive and respectful ways.

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- Engage in problem-solving activities and collaborative exercises.

Consistent participation will improve both your understanding of the material and your performance in assessments.

Effat University's Deanship of Student Affairs provides a variety of dynamic and flexible opportunities through the Ambassador Program and Student Life initiatives. Students are encouraged to actively engage in these activities, which foster leadership skills, teamwork, and a strong sense of community. Please refer to their announcements for guidance on fulfilling your Student Development & Participation requirements this semester.

Course Description

This course provides an in-depth introduction to neural networks and deep learning. Students will learn key methods for improving neural networks, including hyperparameter tuning, regularization, and optimization. Core topics include Convolutional Neural Networks (CNNs) for image tasks, sequence models for time-series and language, and generative models for data creation. Advanced topics cover recent developments in deep learning, including attention mechanisms, transformers, Large Language Models (LLMs), transfer learning, and ethical considerations. Through hands-on laboratories and projects, students will develop practical skills to build, optimize, evaluate, and apply deep learning models to real-world AI applications.

Prescribed Textbooks:

- Aggarwal, C. C. (2023). Neural Networks and Deep Learning: A Textbook. Springer.

Link to Effat University Library Website:

- <http://library.effatuniversity.edu.sa/>

You can also visit the library or use email library@effatuniversity.edu.sa to help you get an e-copy of your textbook.

Additional Recommended Readings:

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
- Howard, J., & Gugger, S. (2020). Deep Learning for Coders with fastai and PyTorch: AI Applications Without a PhD. O'Reilly Media.

Research Papers Required:

- He, K., Zhang, X., Ren, S., & Sun, J. (2016). "Deep Residual Learning for Image Recognition." CVPR <https://arxiv.org/abs/1512.03385>
- Vaswani, A., et al. (2017). "Attention Is All You Need." NeurIPS. <https://arxiv.org/abs/1706.03762>

Sustainable Development Goals (SDG) Connection

SDG 9- Industry Innovation and Infrastructure

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Course Outline

Week	Topic	Class Plan	Readings Material	Tasks and Assignments
Week 1	Introduction to the course and background of deep learning and its application	General Discussion Lecture	Chapter 1: An Introduction to Neural Networks	Project overview and introduction to available DL frameworks
Week 2	The Backpropagation Algorithm	Lecture Class Activity	Chapter 2: The Backpropagation Algorithm	Assignment 1 Project and team selection
Week 3	Deep Learning: Training Deep Neural Networks	Class Activity Lab Practice	Chapter 4: Deep Learning: Principles and Training Algorithms	Quiz 1 Lab1
Week 4	Improving Deep Neural Networks: Regularization and Generalization	Lecture & Discussion Lab practice	Chapter 5: Teaching Deep Learners to Generalize	Assignment 2 Lab2
Week 5	Optimization and Hyperparameter Tuning	Lecture & Problem Solving Lab practice	Chapter 4 — selected sections on training algorithms + Chapter 5 — selected sections	Quiz 2 Project Part 1 Due
Week 6	Convolutional Neural Networks I: Foundations	Lecture & Problem Solving Lab practice	Chapter 9: Convolutional Neural Networks — first part	Project
Week 7	Midterm Exams			Midterm
Week 8	Convolutional Neural Networks II: Architectures and Transfer Learning	Lecture & Problem Solving Lab practice Testing and debugging	Chapter 9: Convolutional Neural Networks — remaining selected sections. Supplement: He et al. (2016), Deep Residual Learning for Image Recognition	Assignment 3 Lab3
Week 9	Recurrent Neural Networks I: Modeling Sequences	Lecture & Problem Solving & Discussion Lab practice Testing and debugging	Chapter 8: Recurrent Neural Networks — RNN foundations	Quiz 3
Week 10	Recurrent Neural Networks II: LSTM and GRU	Lecture & Problem Solving & Discussion Lab practice	Chapter 8: Recurrent Neural Networks — LSTM/GRU sections	Assignment 4 Lab4
Week 11	Attention Mechanisms	Lecture & Problem Solving & Discussion Lab practice	Chapter 12: Advanced Topics in Deep Learning — selected relevant sections + instructor notes. Supplement: Vaswani et al. (2017), Attention Is All You Need — selected sections	Quiz 4

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Week 12	Transformers	Lecture & Problem Solving & Discussion	Chapter 12: Advanced Topics in Deep Learning	Assignment 5
Week 13	Large Language Models and Transfer Learning	Lecture & Problem Solving & Discussion	Chapter 12: Advanced Topics in Deep Learning	Lab5
Week 14	Generative Deep Learning and GANs	Lecture & Problem Solving & Discussion	Chapter 12: Advanced Topics in Deep Learning	
Week 15	Final Project Presentations			Project Presentation
Weeks 16,17	Final Exam			

Assessment Strategy and Course Learning Outcomes

Assessment in this course is designed to measure your achievement of the Course Learning Outcomes (CLOs) and to support your progress throughout the semester. It includes both formative assessment (ongoing feedback during the learning process) and summative assessment (final evaluation at the end of a topic or course).

Effat University's IQRA-based Overarching Assessment Strategy provides detailed guidelines on aligning assessment with learning outcomes. Faculty refer to the assessment manuals in their academic programs in designing this semester's strategies.

Effat Graduate Characteristics	NQF	Aligned PLOs	Course Learning Outcomes	Assessment Tool
Knowledge and Understanding				
Itqan الإتقان	K1	K1 (PLO-7) (ABET 7)	CLO1: Demonstrate understanding of key deep learning concepts, architectures, and techniques.	FinalQ1, MidtermQ1, QZ1
	K2	K2 (PLO-7) (ABET 7)	CLO2: Explain the fundamentals of Convolutional Neural Networks (CNNs) and sequence models.	FinalQ2, MidtermQ2, QZ2
Skills				
Cognitive Skills				
Itqan الإتقان	S1.1			
	S1.2			
	S3.4			
Practical and Physical Skills				
Itqan الإتقان	S2.1			
	S2.2			
Communication and IT Skills				
Ambassador السفيرة	S3.1	S3(PLO-1) (ABET 1)	CLO3: Build deep learning models using modern frameworks.	FinalQ3, MidtermQ3, QZ3, Lab1,Lab2
Itqan الإتقان	S3.2		CLO4: Implement various deep learning techniques for different types of data.	FinalQ4, QZ4, Lab3,Lab4

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Effat Graduate Characteristics	NQF	Aligned PLOs	Course Learning Outcomes	Assessment Tool
	S3.3		CLO5: Evaluate deep learning models performance.	Project, Lab5
Values, Autonomy, and Responsibility				
Values and Ethics				
Ihsan الإحسان	V1			
Autonomy and Responsibility				
Stewardship الراعية	V2	V2 (PLO-4) (ABET 4)	CLO6: Collaborate with peers to solve real-world challenges using deep learning.	Project
	V3			
	V4			

Grade Distribution

[All course-graded work should be listed here. The grade distribution should follow the Ministry of Education bylaws, whereby the Midterm exam counts for 20% and the Final Exam (Practical and theoretical) counts for 40% for all courses except Research-based courses or capstone courses. Any deviation from this grading system should be discussed and approved in DC and CC and endorsed by the provost]

Type of Assessment

Grading

Midterm Exam	20%
Final Exam	20%
Attendance and Participation	5%
Student Development & Participation	5%
Team Project & Research Project	20%
Homework/ Assignment or other	10%
Quizzes	10%
Labs	10%

Delivery of Assessment

- All assignments will be posted on Blackboard with detailed submission guidelines and an evaluation rubric.
- Quizzes, midterm exams, and final exams will be held on campus.

Submission of Assessment

- Assessment that requires in-person submission should be submitted as per the faculty's requirement.
- Missing the submission deadline, quiz, or exam will result in losing full marks of the assessment.

Evaluation of Assessment

- This course will utilize online tools designed to facilitate fair evaluation of student work and its integrity such as Turnitin, Grammarly, and Blackboard SafeAssign.
- Faculty will use the evaluation rubrics for grading each assessment.
- Grades and feedback on student-submitted work will be posted on Blackboard Gradebook within one week of the assessment submission date.

Assessment Details and Rubrics

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[Every course should include team project and/or Research activity stated clearly in the Grade Distribution. The grading activities should be described in detail in the syllabus]

Criteria	Excellent	Good	Fair	Poor
Technical Implementation	Correct, complete, well-organized implementation	Mostly functional; minor issues	Partially functional; several errors	Incomplete or non-functional
Model Design & Performance	Well-justified architecture; strong performance and appropriate metrics	Appropriate design; reasonable performance	Limited justification; weak/moderate performance	Inappropriate design or no valid evaluation
Innovation & Relevance	Original, well-motivated and highly relevant solution	Relevant with some originality	Relevant but limited originality/depth	Weak relevance or motivation
Experimental Analysis	Thorough analysis, comparisons and critical interpretation	Good analysis with some limitations discussed	Limited interpretation or comparison	Little/no meaningful analysis
Presentation & Report	Clear, professional and technically accurate	Clear with minor issues	Somewhat unclear or poorly organized	Incomplete or difficult to follow

Additional Enrichment and Learning Activities

To further enhance the learning experience in this course, the following enrichment components may be included (as determined by the instructor and department):

- Coursera and/or LinkedIn Learning Courses**
 - Students may be required to complete selected online modules that reinforce course concepts.
 - Certificates of completion can be uploaded to Blackboard as part of continuous assessment.
- Professional Certificates (if applicable)**
 - Where relevant, students may pursue professional certifications linked to course outcomes (e.g., software, technical, or industry-recognized certifications).
 - Faculty will clarify requirements and deadlines.
- Guest Speakers and Tentative Lecture Titles**
 - Invited experts from academia or industry may deliver guest lectures on specialized topics.
 - Tentative lecture titles will be shared with students in advance.
- Service-Learning Activities (if applicable)**
 - Students may participate in community-oriented projects that connect course content to real-world applications and align with Effat University's mission and the SDGs.

- Guest Speakers: AI

Guidelines for AI Usage in the Course

The use of generative AI tools (e.g., ChatGPT, DALL·E, etc.) in this course is permitted **only if approved by the faculty**. When approval is granted, AI tools may be used as a supplemental resource to support your learning, and the following guidelines must be followed:

- Do not violate intellectual property laws.
- Do not include misinformation or unethical content.

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- Clearly distinguish between your own work and AI-generated material.
- AI-generated content must not exceed 25% of any submitted work.
- Always provide proper citation when AI tools are used.
- Use proper authorization and citation. Failure to cite your sources will be considered plagiarism and treated as a violation of Effat University's Code of Conduct and Tarbawyyat guidelines.

Academic Integrity

At Effat University, every member is expected to uphold the highest standards of honesty at all times. Our mission is to promote ethical conduct in the management of academic, research, and community service activities, as well as in all interactions with our stakeholders. *Tarbawyyat Effat* outlines eight fundamental pillars that form the foundation of our code of ethical conduct:

1. تقوى (Taqwā) – Piety
2. آداب (Ādab) – Knowledge, Manners, and Outreach
3. رعاية (Ri'ayya) – Nurturing
4. بنيان مرصوص (Bunyan Marsūs) – Cooperation and Collaboration
5. وسطية (Wasatya) – Tolerance and Moderation
6. يسر (Yusr) – Modesty and Ease
7. أمانة (Amanah) – Ethics and Integrity
8. توجيه (Tawḡyḥ) – Guidance

Students, as integral members of the Effat community, share the responsibility of adhering to the principles and spirit of *Tarbawyyat Effat*. Any activity that compromises the fair evaluation of a student's performance is strictly prohibited. Such activities include, but are not limited to, cheating, plagiarism, and taking unfair advantage. All students must comply with *Tarbawyyat Effat* at all times. The full code can be reviewed on the University website. For further details on procedures in cases of violations, refer to **Article 29** of the *Effat University Bylaws for Undergraduate Study and Examination and Implementation Rules*, which provides clear guidance on the University's processes.

Computer Resources

- Access to Effat University computer labs

This Course Syllabus is prepared by:

Name: Dr Naila Marir

Position: Assistant Professor

Department: Computer Science Department

Semester: Fall 2026